SHUBHAM SONKAR

Electronics engineering student | Embedded System and IoT | Schematic and PCB designer
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PROFILE

Electronics Engineering undergraduate with hands-on experience in PCB design, embedded systems and IoT. Comfortable taking hardware from schematic to working prototype — designing layouts in Altium Designer, programming Arduino and ESP-based boards, and simulating circuits in Proteus. Interned at Hindustan Aeronautics Limited (HAL) and completed a four-week IoT and Embedded Systems internship.

EDUCATION

B.Tech Electronics Engineering(Pursuing) – Current SGPA-7.4
Institute of Engineering and Rural Technology (AKTU)

Intermediate (CISCE) - 85.5%
St. Thomas Mission School, Lucknow

High School (CISCE) - 75.5%
St. Thomas Mission School, Lucknow

TECHNICAL SKILLS & COMPETENCIES

Core: Circuit Design · PCB Design & Layout · Embedded Systems & IoT · Schematic Design

Software & Tools: Altium Designer · Xilinx Vivado · Proteus · Keysight ADS · Arduino IDE · VS Code · MS Word · MS Excel · MS PowerPoint

Soft Skills: Communication, Team Collaboration, Creative Problem Solving, Time Management, Attention to Detail

INTERNSHIPS

Hindustan Aeronautics Limited (HAL), Lucknow Jun 2026 – Jul 2026
Industrial Internship

- Studied aircraft instrumentation and control applications, including the fuel-management system of the SU-30 aircraft.
- Gained practical exposure to sensors, measurement systems and electronic control used in aerospace manufacturing.

Upskill Campus (with UniConverge Technologies Pvt. Ltd.) Jul 2026 – Aug 2026
IoT & Embedded Systems Internship

- Completed a four-week internship working with Arduino programming, sensors and hardware interfacing.
- Built a Temperature and Humidity Monitoring System using an Arduino Nano and DHT11 sensor — simulated in Proteus, programmed in the Arduino IDE, and displayed live on a 16×2 LCD with relay and buzzer threshold alerts.

CERTIFICATIONS

- VLSI Design – Simplilearn Skill up
- PCB Basic Design – Altium Education
- Digital Electronics – Maven Silicon
- Python Programming – Simplilearn Skill up
- PLC Course – Automation Community
- C++ – Simplilearn Skill up

LANGUAGES

English (Professional) | Hindi (Native)

PROJECTS

Custom ESP8266 Development Board
Sep 2026

PCB Design

- Designed a custom ESP8266 development board in Altium Designer with CH340C USB-to-UART programming, 12 V input power management, regulated 5 V / 3.3 V rails, status indicators and GPIO headers.
- Developed the schematic, footprints, PCB placement, routing and 3D board layout, applying signal- and power-routing best practices.

I²C Multi-Node Communication System
Sep 2026

PCB Design

- Designed an I²C master–slave system using ESP32 and Arduino Nano with 3.3 V–5 V level shifting, SDA/SCL pull-up networks and multi-node communication for sensor and control applications.

Distributed Safety Surveillance & Response Network
Aug 2026

Embedded Monitoring & Control

- Built a distributed monitoring system using ESP32-S3 and Arduino Nano with DHT11, MQ-5, flame, LDR and PIR sensors, including I²C communication, relay-based response control and Wi-Fi connectivity for remote monitoring.

Temperature and Humidity Monitoring System

Embedded Monitoring & Control Jun 2026

- Arduino Nano based system that monitors temperature and humidity in real time using a DHT11 sensor, shown on a 16×2 LCD with a power-on indicator.
- When temperature crosses a preset threshold, a relay switches on a DC fan, a buzzer sounds an alert and status LEDs indicate the system state.

RF Filter Design
Dec 2025

Keysight Advanced Design System (ADS)

- Designed and simulated an RF filter in Keysight ADS, analysing frequency response, insertion loss and return loss to apply core RF and microwave design principles.